Chapter 50

An Introduction to Ecology and the Biosphere

PowerPoint Lectures for
Biology, Seventh Edition
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Lectures by Chris Romero

- Overview: The Scope of Ecology
- Ecology
  - Is the scientific study of the interactions between organisms and the environment
- These interactions
  - Determine both the distribution of organisms and their abundance

- Ecology
  - Is an enormously complex and exciting area of biology
  - Reveals the richness of the biosphere

Figure 50.1

- Concept 50.1: Ecology is the study of interactions between organisms and the environment
- Ecology
  - Has a long history as a descriptive science
  - Is also a rigorous experimental science

Ecology and Evolutionary Biology

- Events that occur in ecological time
  - Affect life on the scale of evolutionary time

Organisms and the Environment

- The environment of any organism includes
  - Abiotic, or nonliving components
  - Biotic, or living components
  - All the organisms living in the environment, the biota

- Environmental components
  - Affect the distribution and abundance of organisms

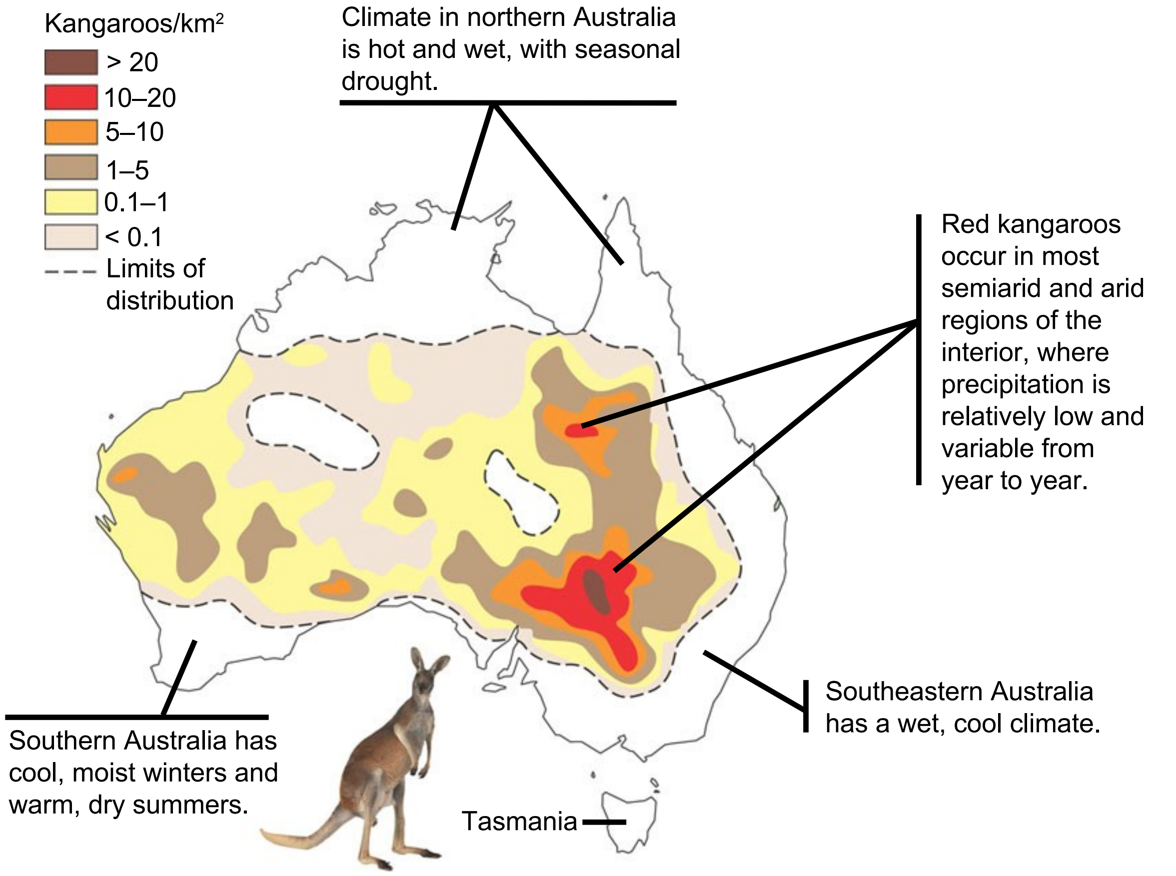


Figure 50.2

- Ecologists
  - Use observations and experiments to test explanations for the distribution and abundance of species

Subfields of Ecology

- Organismal ecology
  - Studies how an organism's structure, physiology, and (for animals) behavior meet the challenges posed by the environment

Figure 50.3a

(a) Organismal ecology. How do humpback whales select their calving areas?

- Population ecology
  - Concentrates mainly on factors that affect how many individuals of a particular species live in an area

(b) Population ecology. What environmental factors affect the reproductive rate of deer mice?

Figure 50.3b

- Community ecology
  - Deals with the whole array of interacting species in a community

(c) **Community ecology.** What factors influence the diversity of species that make up a particular forest?

Figure 50.3c

- Ecosystem ecology
  - Emphasizes energy flow and chemical cycling among the various biotic and abiotic components

(d) Ecosystem ecology. What factors control photosynthetic productivity in a temperate grassland ecosystem?

Figure 50.3d

- Landscape ecology
  - Deals with arrays of ecosystems and how they are arranged in a geographic region

Figure 50.3e

(e) Landscape ecology. To what extent do the trees lining the drainage channels in this landscape serve as corridors of dispersal for forest animals?

- The biosphere
  - Is the global ecosystem, the sum of all the planet’s ecosystems

Ecology and Environmental Issues

- Ecology
  - Provides the scientific understanding underlying environmental issues
- Rachel Carson
  - Is credited with starting the modern environmental movement

Figure 50.4

- Most ecologists follow the precautionary principle regarding environmental issues
- The precautionary principle
  - Basically states that humans need to be concerned with how their actions affect the environment

- Concept 50.2: Interactions between organisms and the environment limit the distribution of species
- Ecologists
  - Have long recognized global and regional patterns of distribution of organisms within the biosphere

- Many naturalists
  - Began to identify broad patterns of distribution by naming biogeographic realms

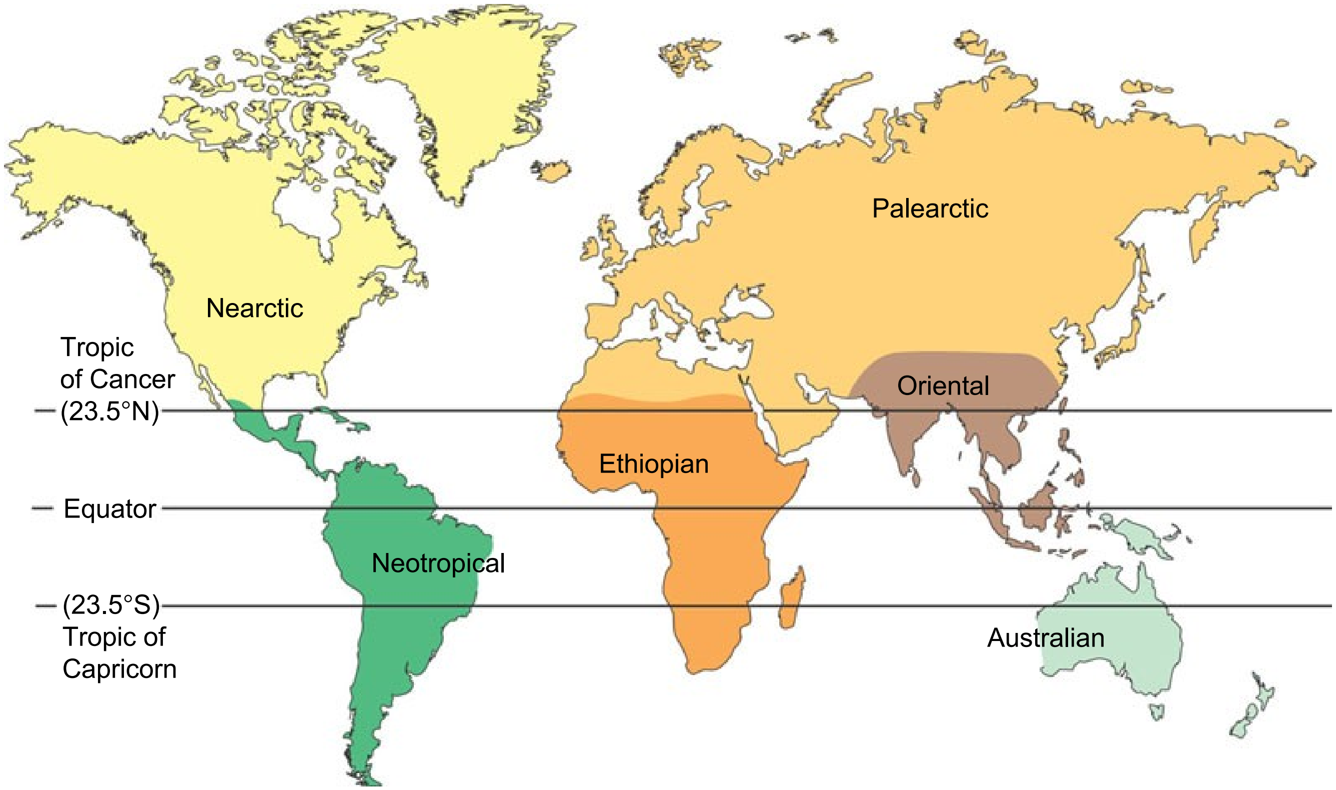


Figure 50.5

- Biogeography
  - Provides a good starting point for understanding what limits the geographic distribution of species

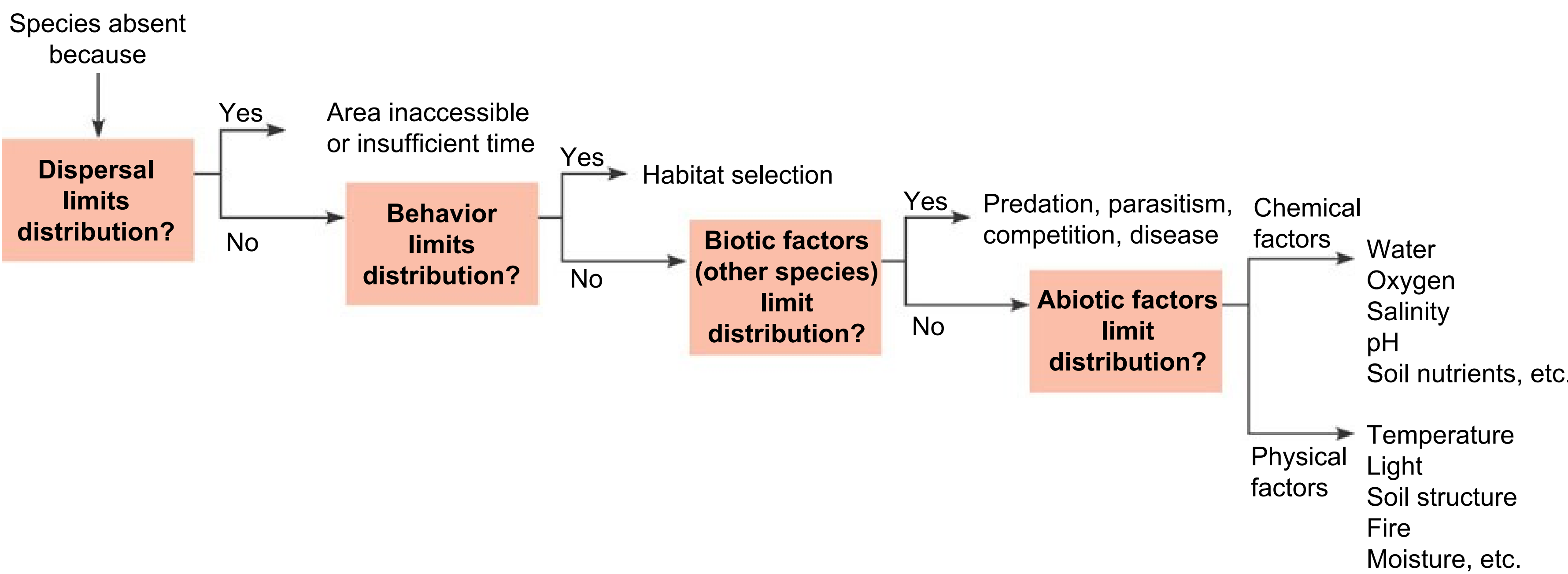


Figure 50.6

Dispersal and Distribution

- Dispersal
  - Is the movement of individuals away from centers of high population density or from their area of origin
  - Contributes to the global distribution of organisms

Natural Range Expansions

- Natural range expansions
  - Show the influence of dispersal on distribution

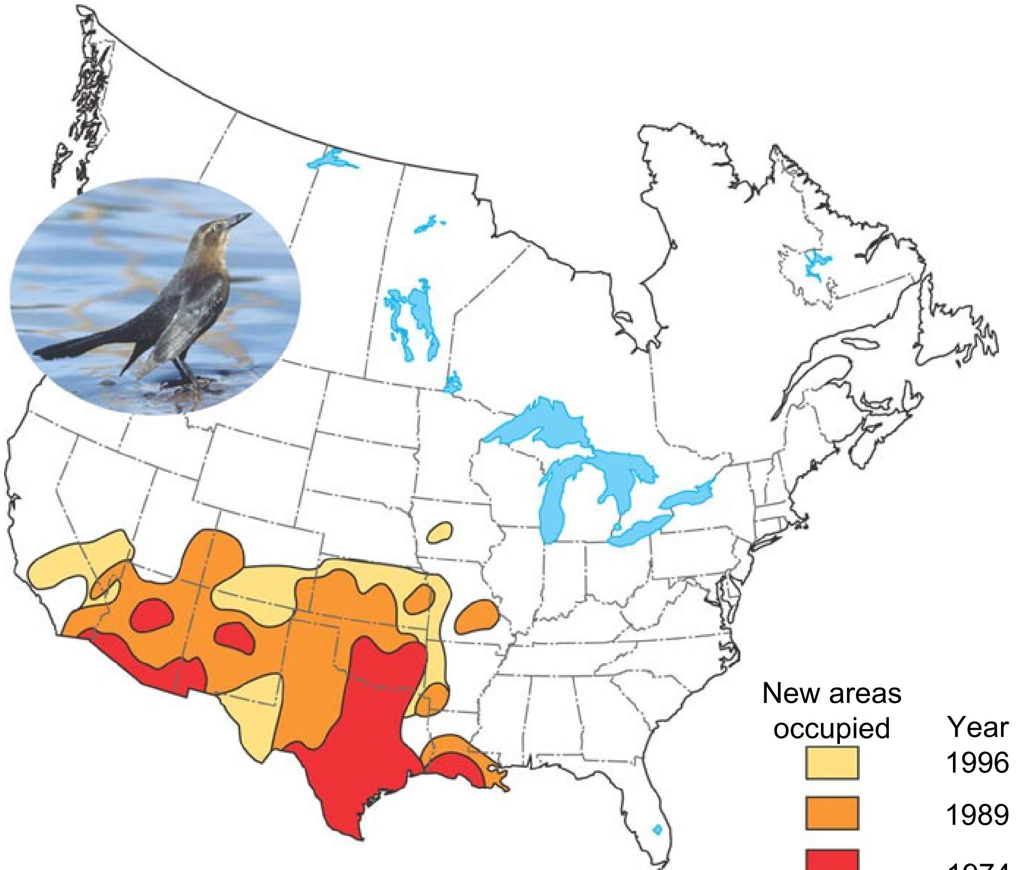


Figure 50.7

Species Transplants

- Species transplants
  - Include organisms that are intentionally or accidentally relocated from their original distribution
  - Can often disrupt the communities or ecosystems to which they have been introduced

Behavior and Habitat Selection

- Some organisms
  - Do not occupy all of their potential range
- Species distribution
  - May be limited by habitat selection behavior

Biotic Factors

- Biotic factors that affect the distribution of organisms may include
  - Interactions with other species
  - Predation
  - Competition

- A specific case of an herbivore limiting distribution of a food species

EXPERIMENT W. J. Fletcher tested the effects of two algae-eating animals, sea urchins and limpets, on seaweed abundance near Sydney, Australia. In areas adjacent to a control site, either the urchins, the limpets, or both were removed.

RESULTS Fletcher observed a large difference in seaweed growth between areas with and without sea urchins.

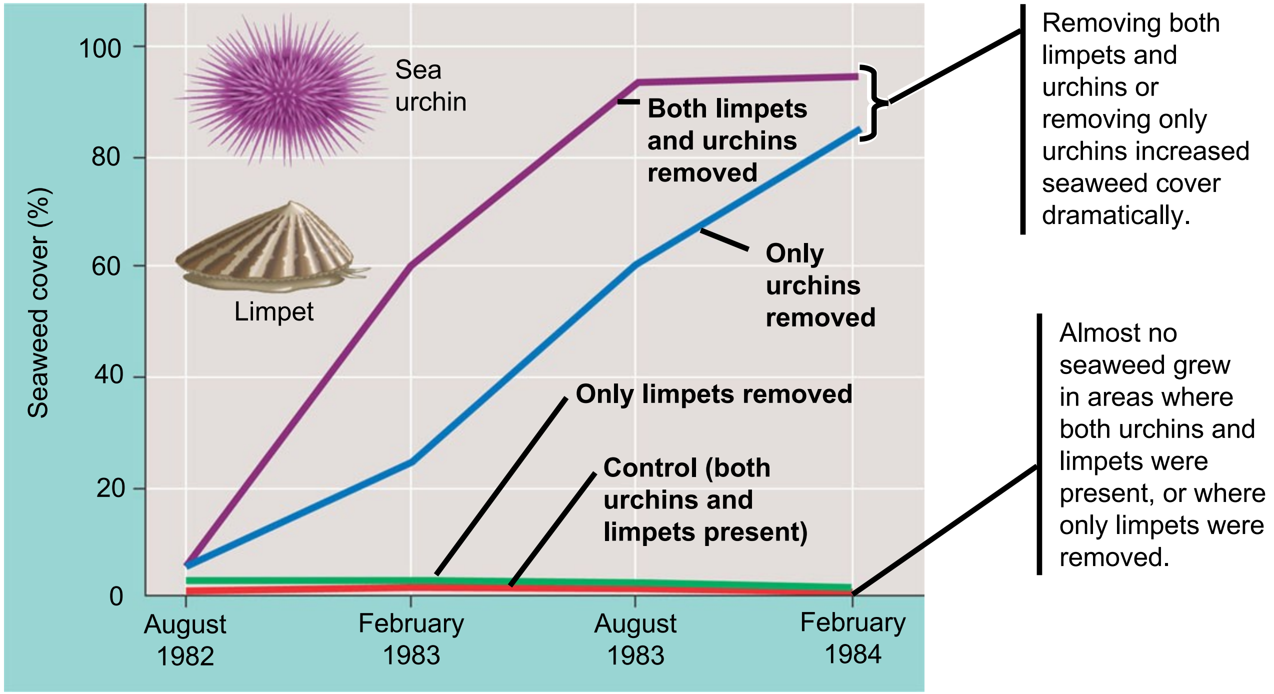


Figure 50.8

CONCLUSION Removing both limpets and urchins resulted in the greatest increase of seaweed cover, indicating that both species have some influence on seaweed distribution. But since removing only urchins greatly increased seaweed growth while removing only limpets had little effect, Fletcher concluded that sea urchins have a much greater effect than limpets in limiting seaweed distribution.

Abiotic Factors

- Abiotic factors that affect the distribution of organisms may include
  - Temperature
  - Water
  - Sunlight
  - Wind
  - Rocks and soil

Temperature

- Environmental temperature
  - Is an important factor in the distribution of organisms because of its effects on biological processes

Water

- Water availability among habitats
  - Is another important factor in species distribution

Sunlight

- Light intensity and quality
  - Can affect photosynthesis in ecosystems
- Light
  - Is also important to the development and behavior of organisms sensitive to the photoperiod

Wind

- Wind
  - Amplifies the effects of temperature on organisms by increasing heat loss due to evaporation and convection
  - Can change the morphology of plants

Figure 50.9

Rocks and Soil

- Many characteristics of soil limit the distribution of plants and thus the animals that feed upon them
  - Physical structure
  - pH
  - Mineral composition

Climate

- Four major abiotic components make up climate
  - Temperature, water, sunlight, and wind
- Climate
  - Is the prevailing weather conditions in a particular area

- Climate patterns can be described on two scales
  - Macroclimate, patterns on the global, regional, and local level
  - Microclimate, very fine patterns, such as those encountered by the community of organisms underneath a fallen log

Global Climate Patterns

- Earth's global climate patterns
  - Are determined largely by the input of solar energy and the planet's movement in space

- Sunlight intensity
  - Plays a major part in determining the Earth's climate patterns

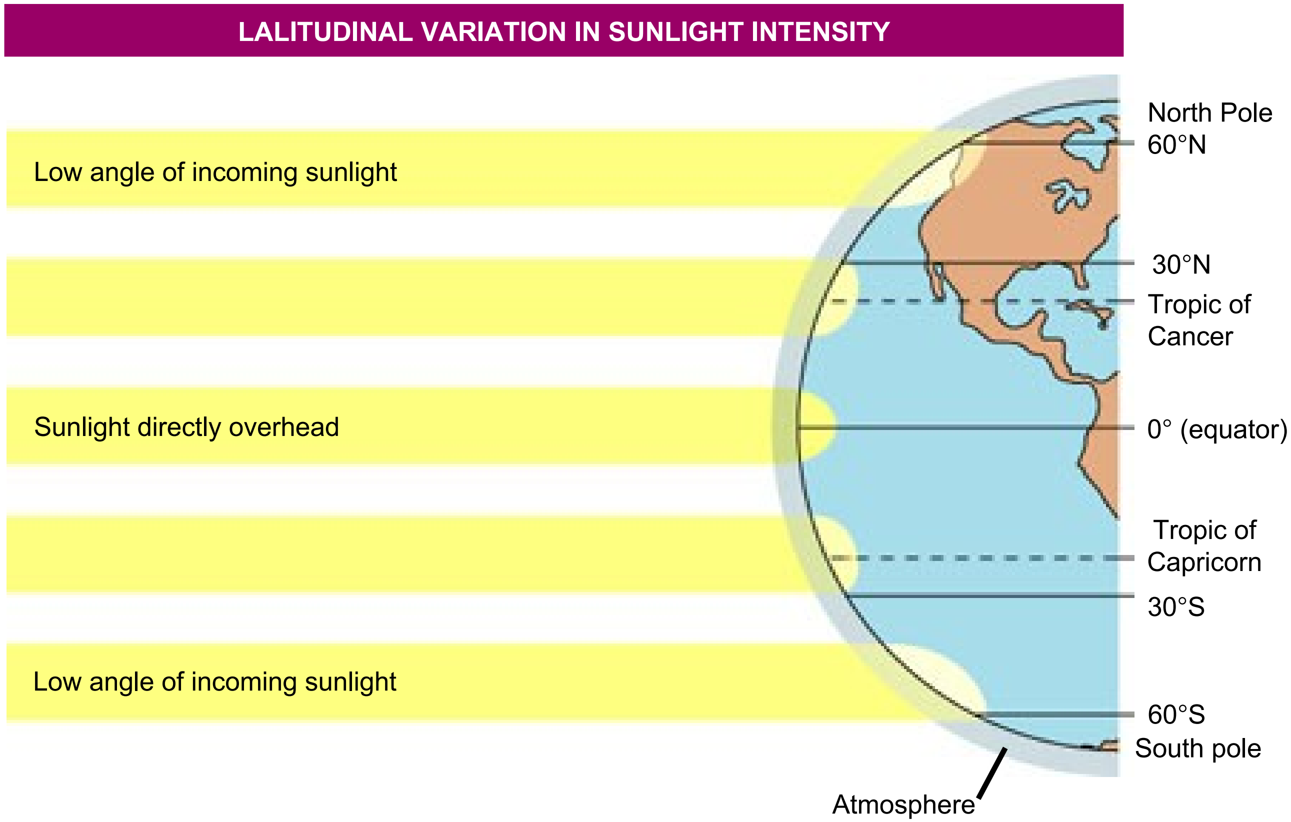


Figure 50.10

SEASONAL VARIATION IN SUNLIGHT INTENSITY

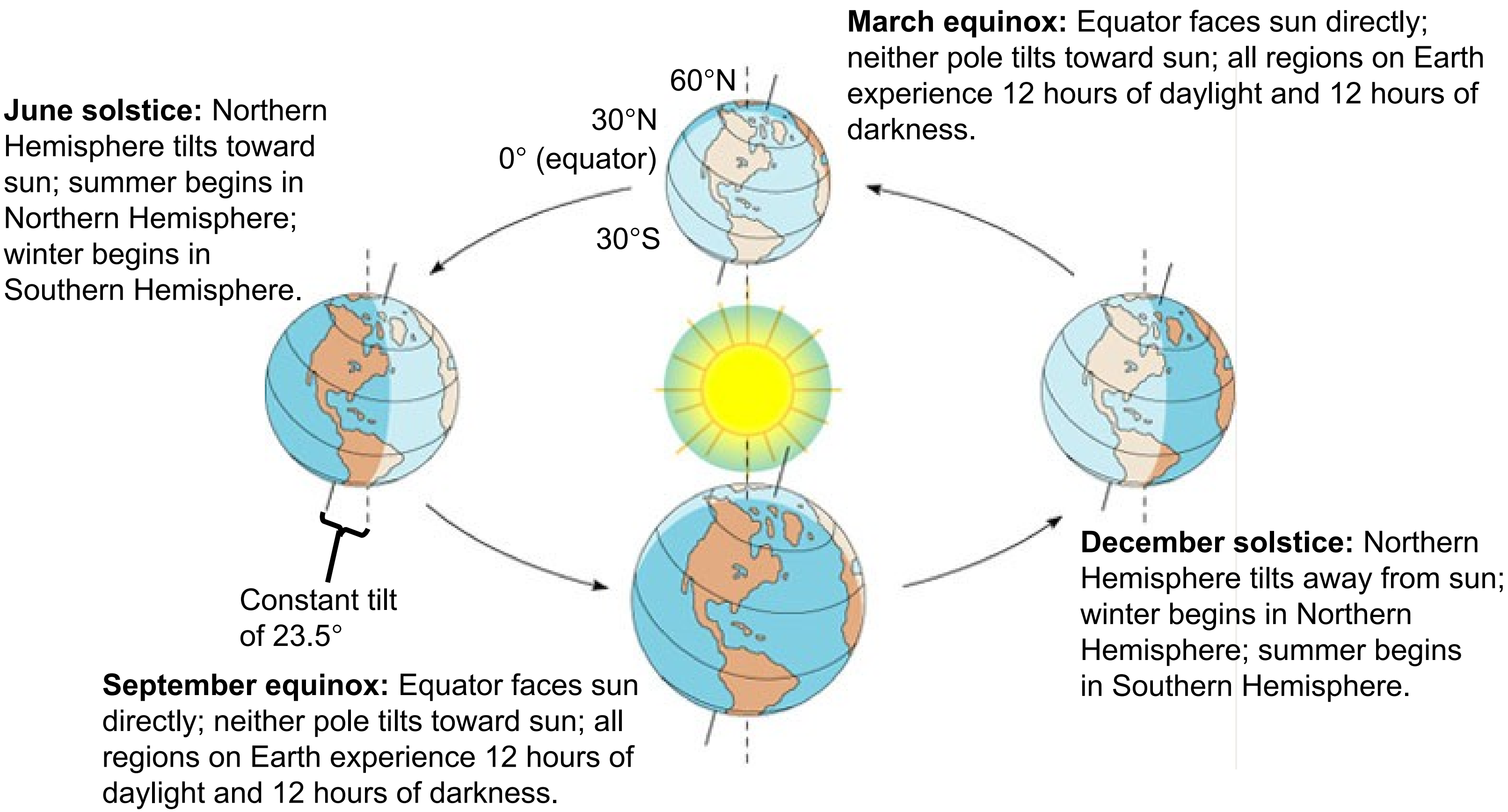


Figure 50.10

- Air circulation and wind patterns
  - Play major parts in determining the Earth's climate patterns

GLOBAL AIR CIRCULATION AND PRECIPITATION PATTERNS

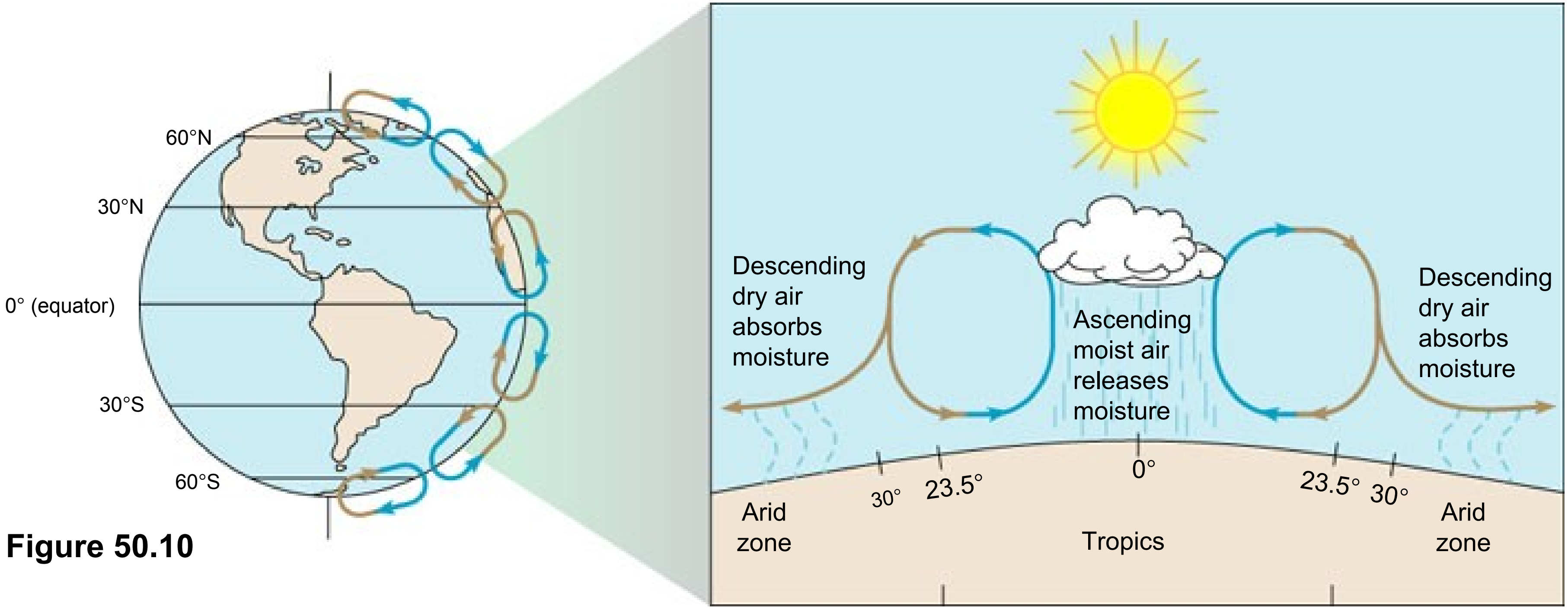


Figure 50.10

GLOBAL WIND PATTERNS

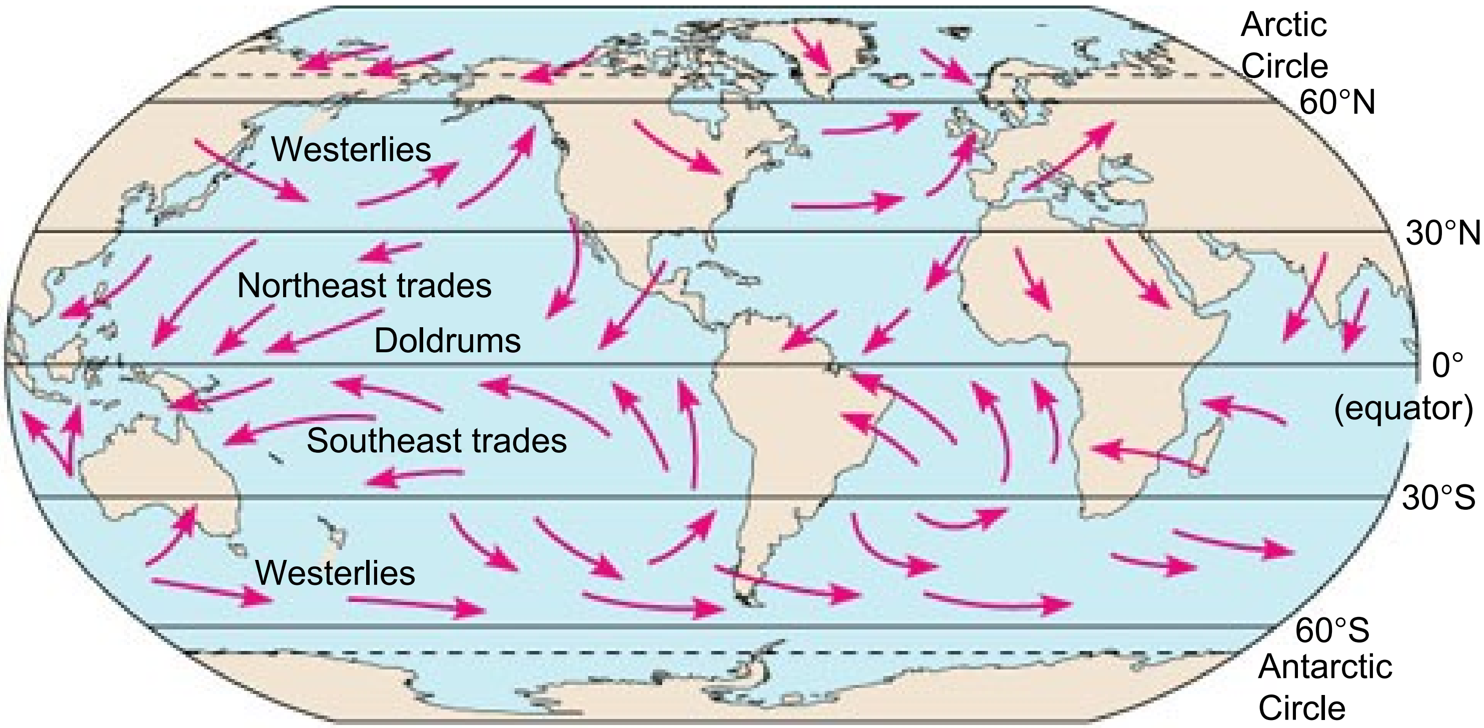


Figure 50.10

Regional, Local, and Seasonal Effects on Climate

- Various features of the landscape
  - Contribute to local variations in climate

Bodies of Water

- Oceans and their currents, and large lakes
  - Moderate the climate of nearby terrestrial environments

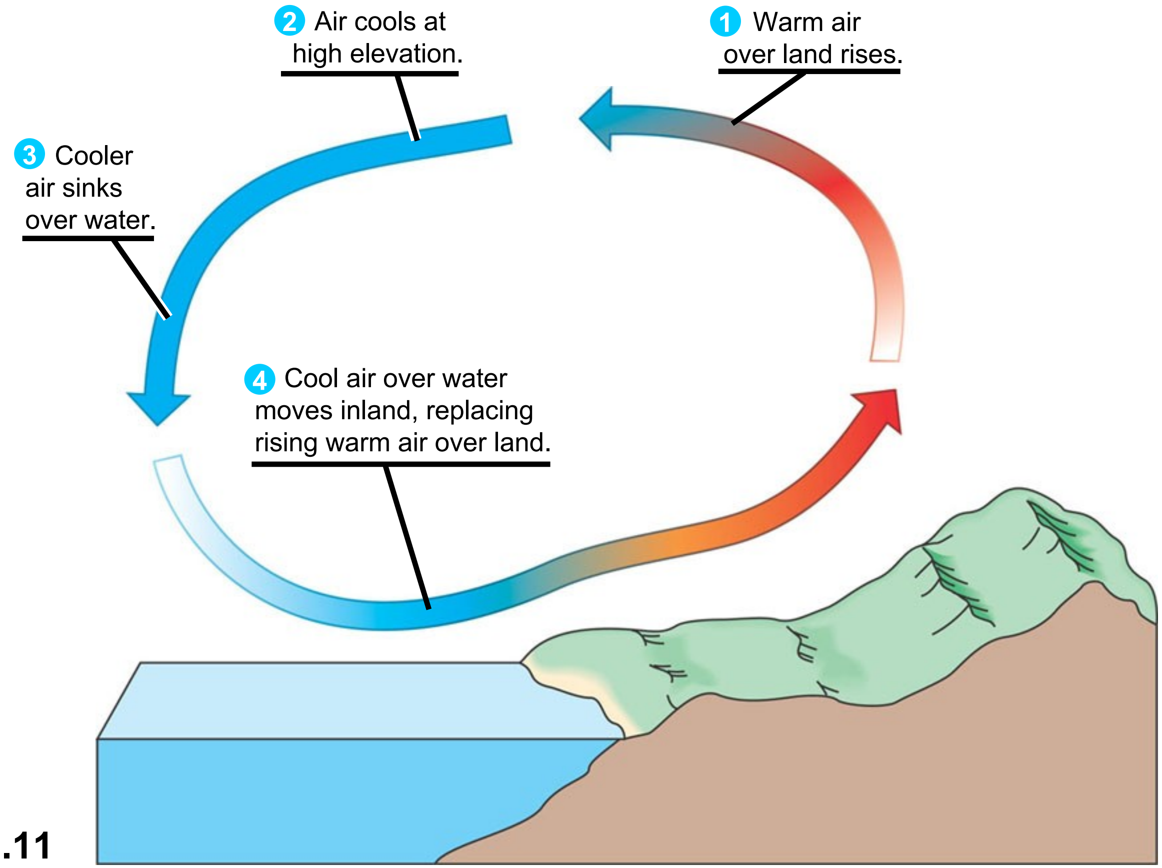


Figure 50.11

Mountains

- Mountains have a significant effect on
  - The amount of sunlight reaching an area
  - Local temperature
  - Rainfall

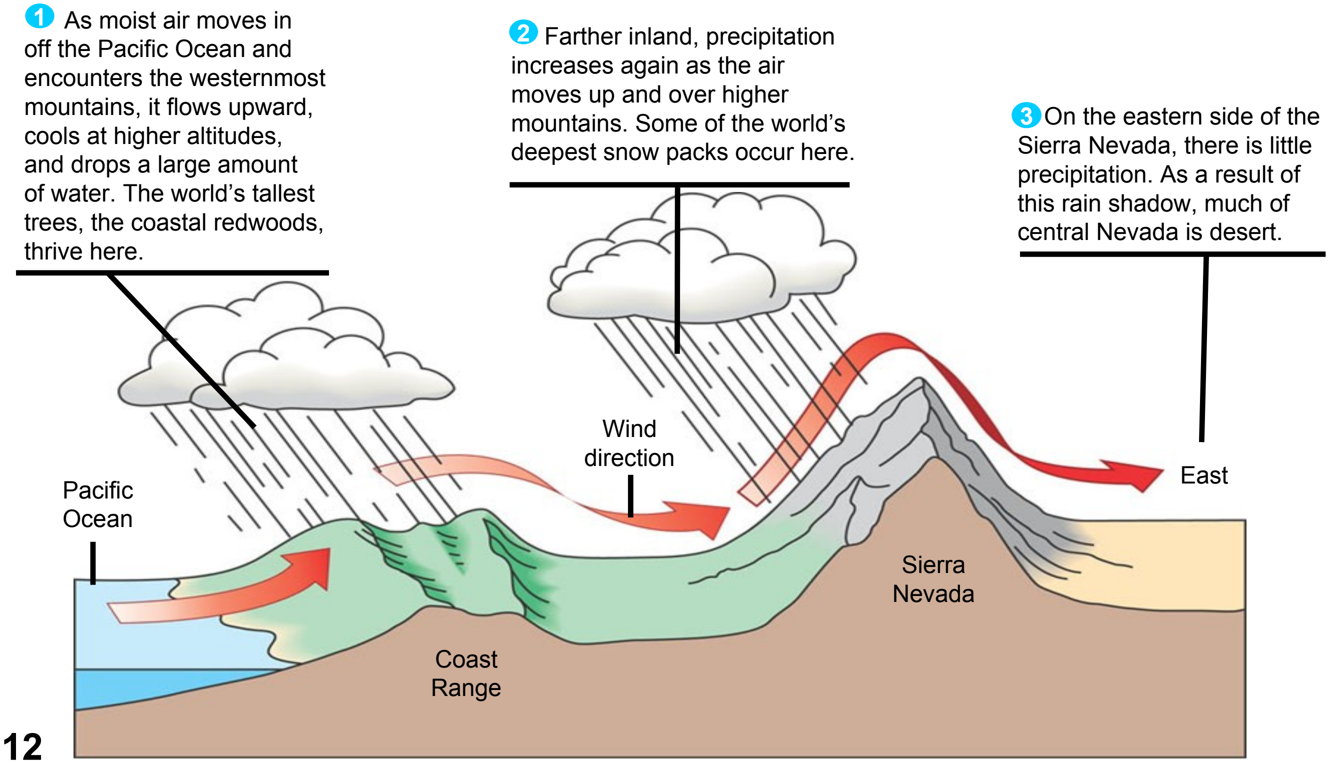


Figure 50.12

Seasonality

- The angle of the sun
  - Leads to many seasonal changes in local environments

- Lakes
  - Are sensitive to seasonal temperature change
  - Experience seasonal turnover

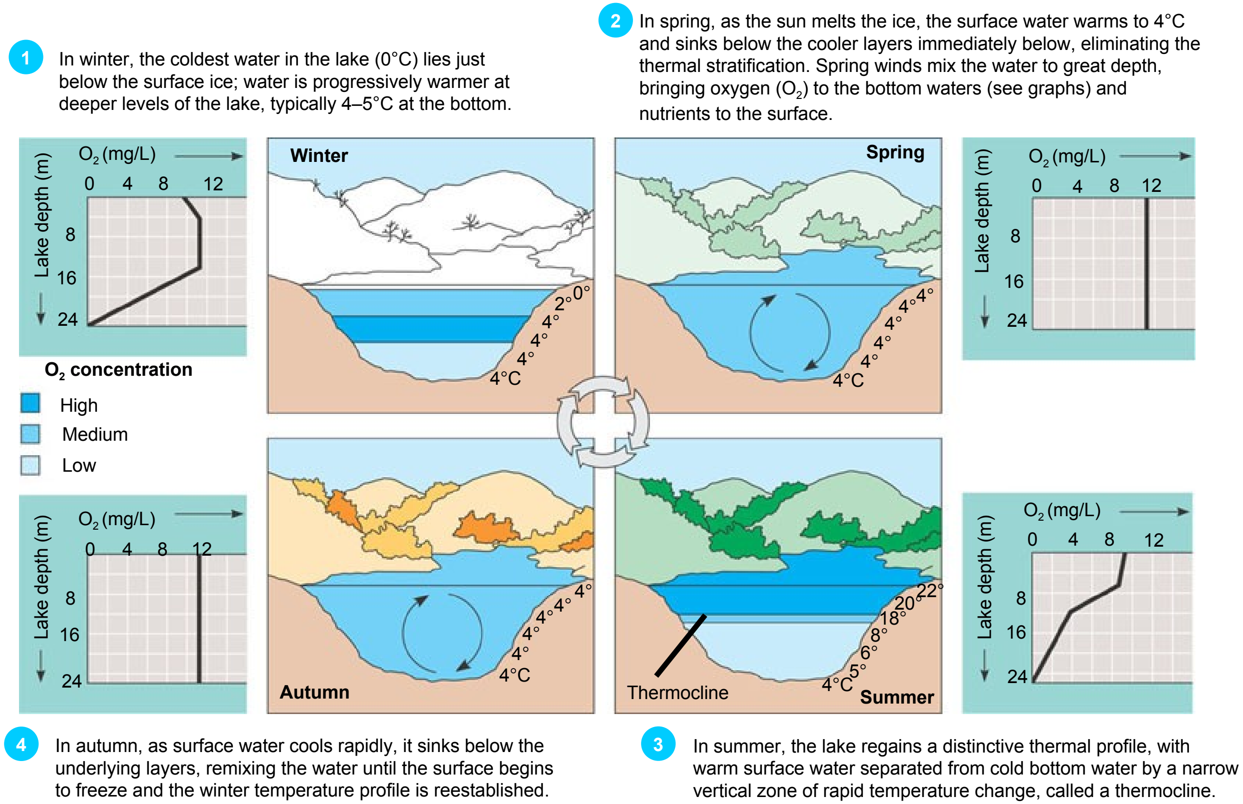


Figure 50.13

Microclimate

- Microclimate
  - Is determined by fine-scale differences in abiotic factors

Long-Term Climate Change

- One way to predict future global climate change
  - Is to look back at the changes that occurred previously

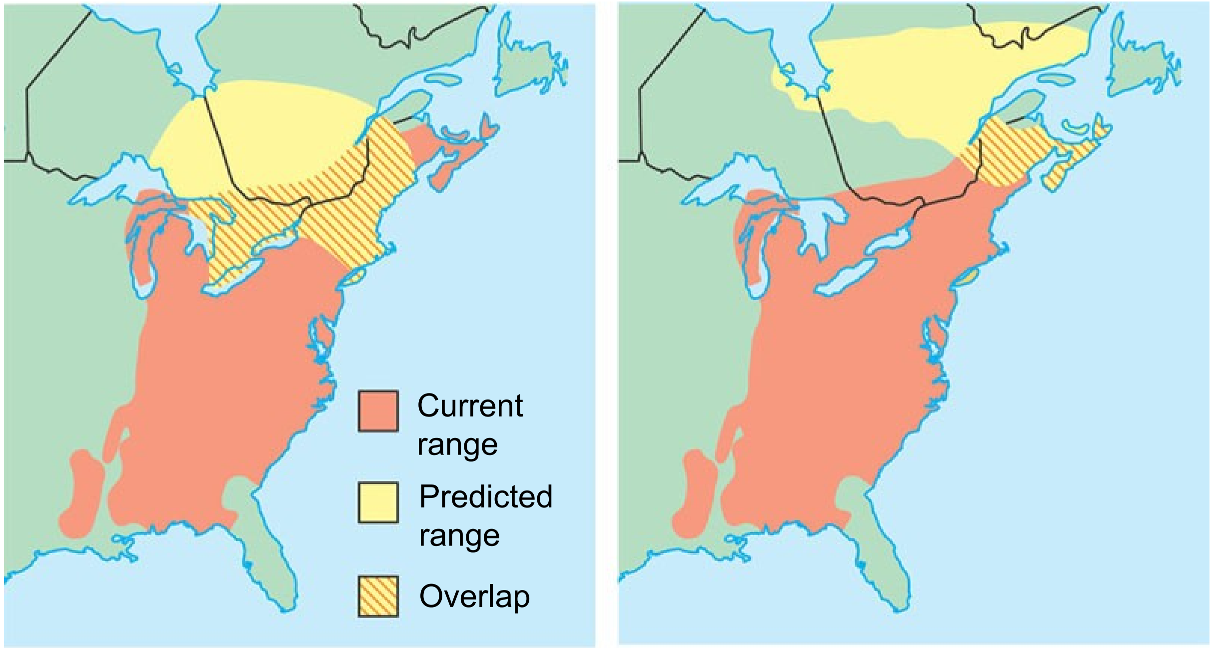


Figure 50.14

(a) 4.5°C warming over next century

(b) 6.5°C warming over next century

- Concept 50.3: Abiotic and biotic factors influence the structure and dynamics of aquatic biomes

- Varying combinations of both biotic and abiotic factors
  - Determine the nature of Earth's many biomes

- Biomes
  - Are the major types of ecological associations that occupy broad geographic regions of land or water

- The examination of biomes will begin with Earth's aquatic biomes

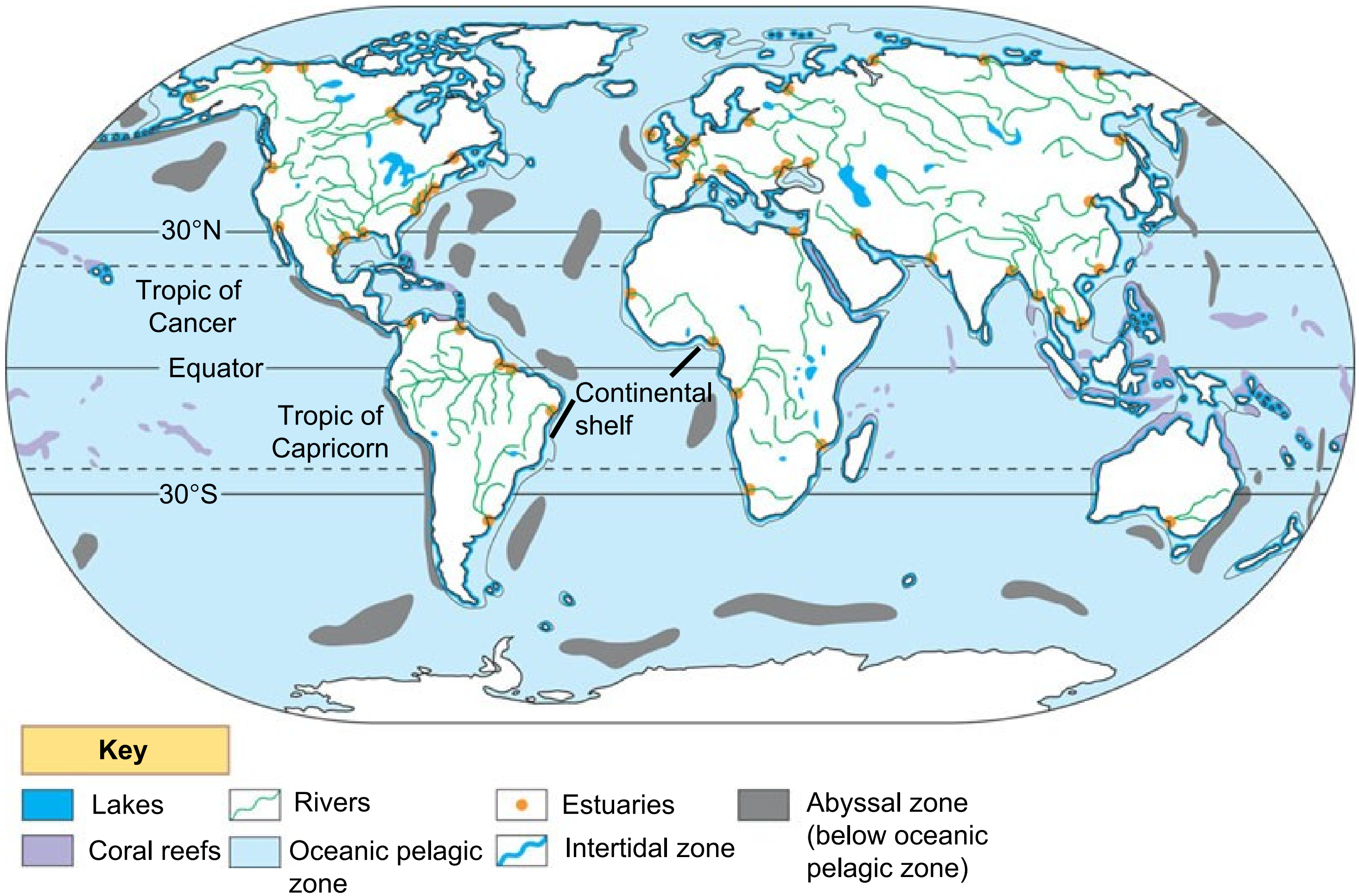
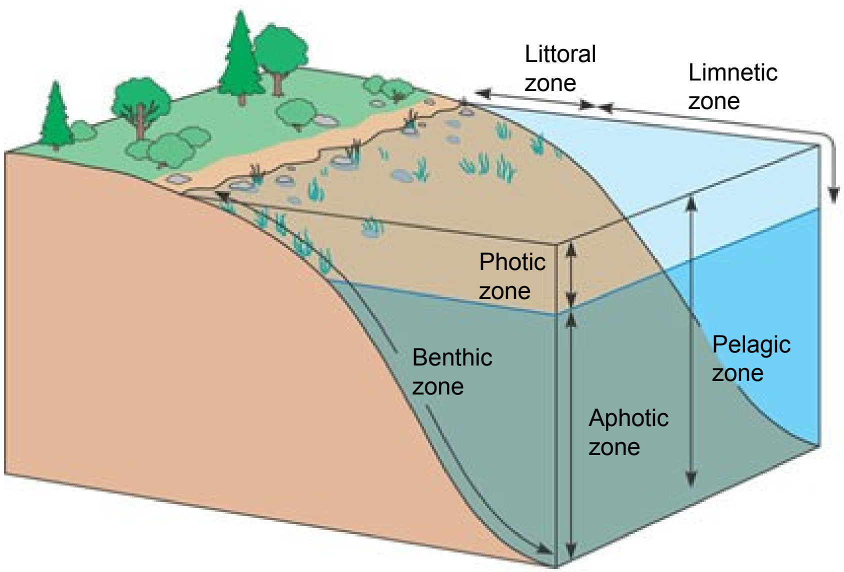


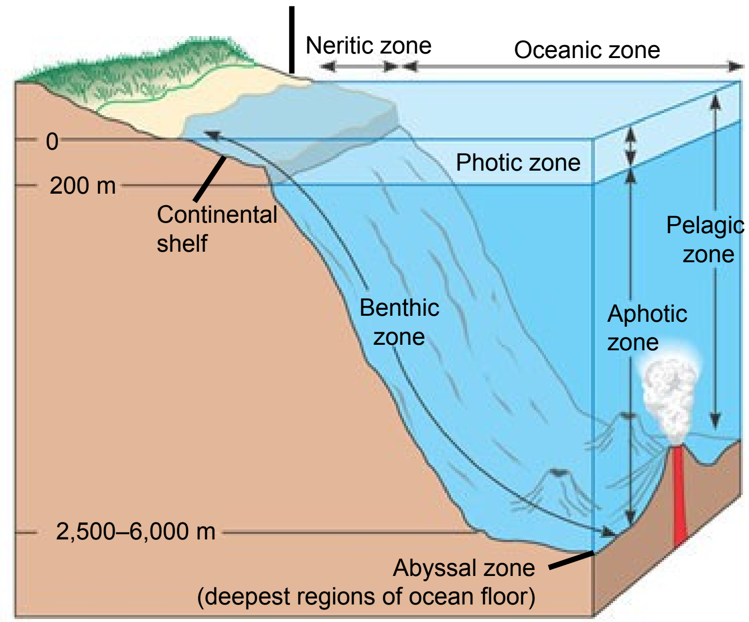
Figure 50.15

- Aquatic biomes
  - Account for the largest part of the biosphere in terms of area
  - Can contain fresh or salt water
- Oceans
  - Cover about 75% of Earth's surface
  - Have an enormous impact on the biosphere

- Many aquatic biomes
  - Are stratified into zones or layers defined by light penetration, temperature, and depth



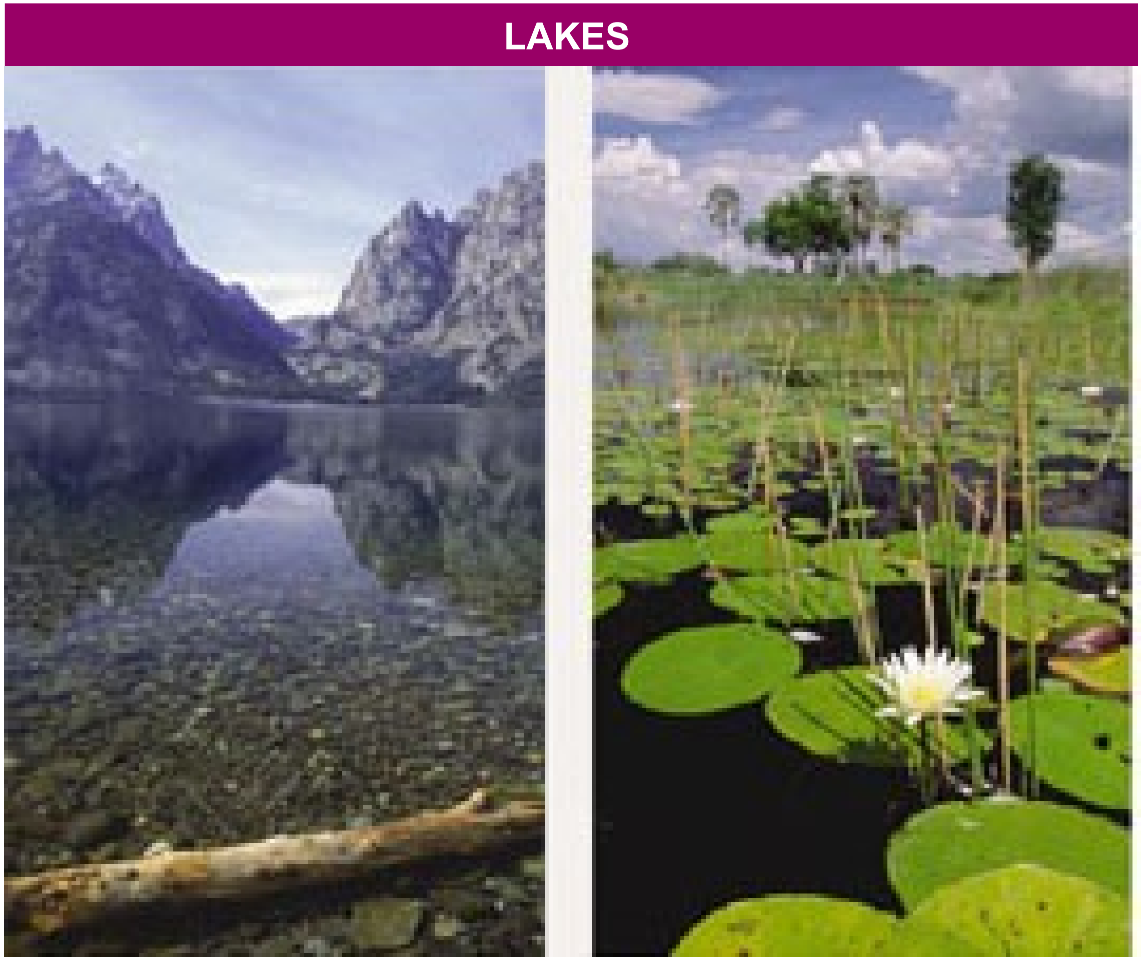
(a) Zonation in a lake. The lake environment is generally classified on the basis of three physical criteria: light penetration (photic and aphotic zones), distance from shore and water depth (littoral and limnetic zones), and whether it is open water (pelagic zone) or bottom (benthic zone).



(b) Marine zonation. Like lakes, the marine environment is generally classified on the basis of light penetration (photic and aphotic zones), distance from shore and water depth (intertidal, neritic, and oceanic zones), and whether it is open water (pelagic zone) or bottom (benthic and abyssal zones).

Figure 50.16a, b

- Lakes



An oligotrophic lake in Grand Teton, Wyoming

A eutrophic lake in Okavango delta, Botswana

Figure 50.17

- Wetlands

Figure 50.17 Okefenokee National Wetland Reserve in Georgia

- Streams and rivers

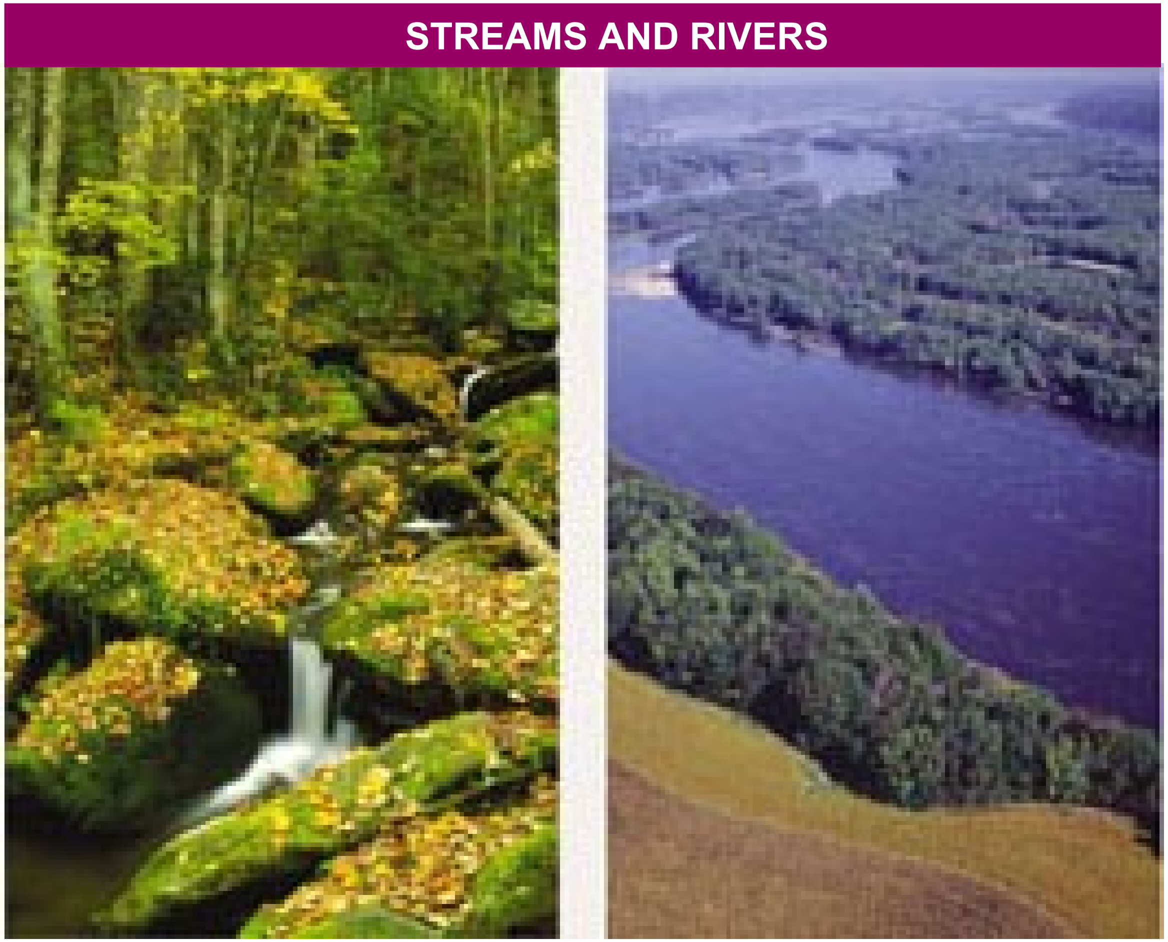


Figure 50.17

A headwater stream in the Great Smoky Mountains

The Mississippi River far form its headwaters

- Estuaries

Figure 50.17 An estuary in a low coastal plain of Georgia

- Intertidal zones

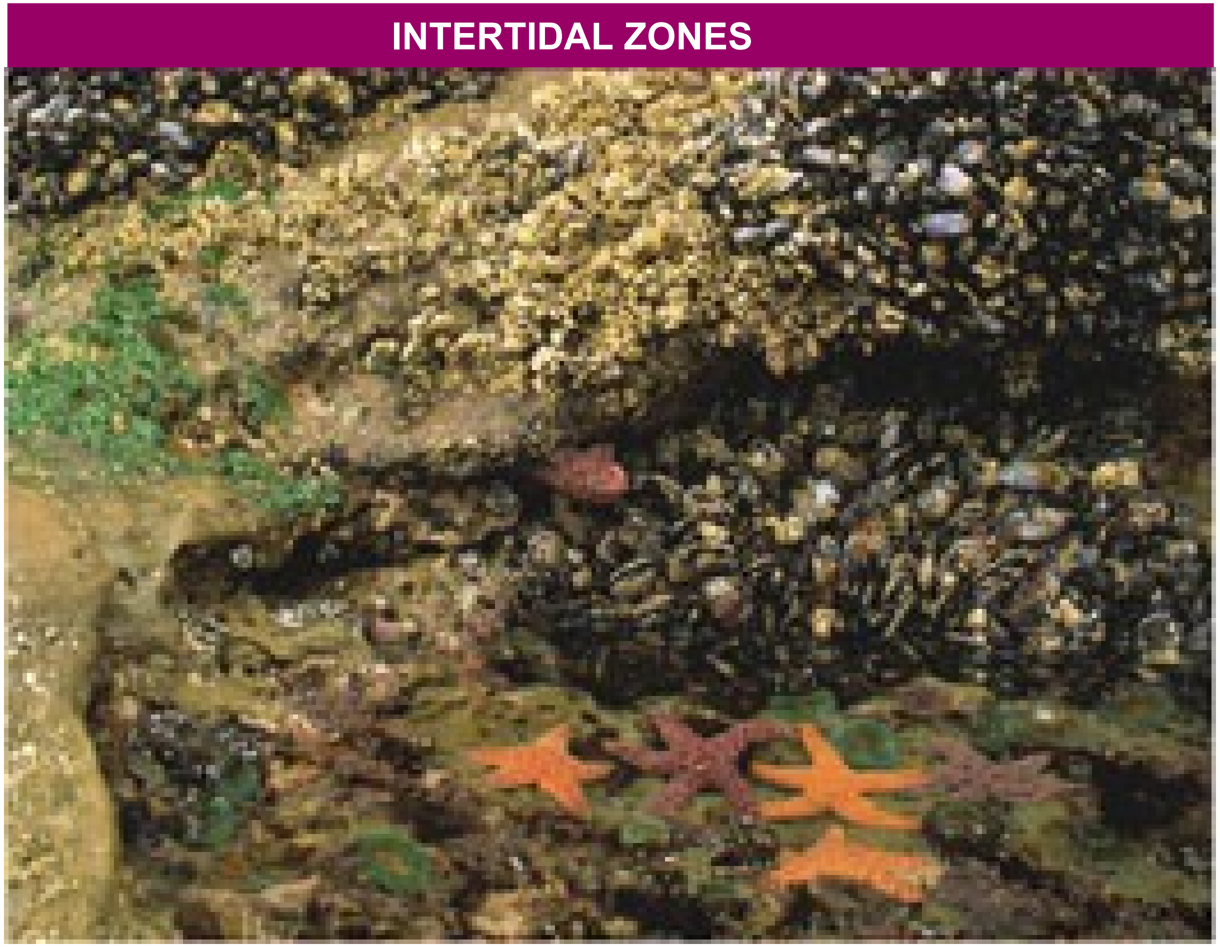


Figure 50.17 Rocky intertidal zone on the Oregon coast

- Oceanic pelagic biome

Figure 50.17 Open ocean off the island of Hawaii

- Coral reefs

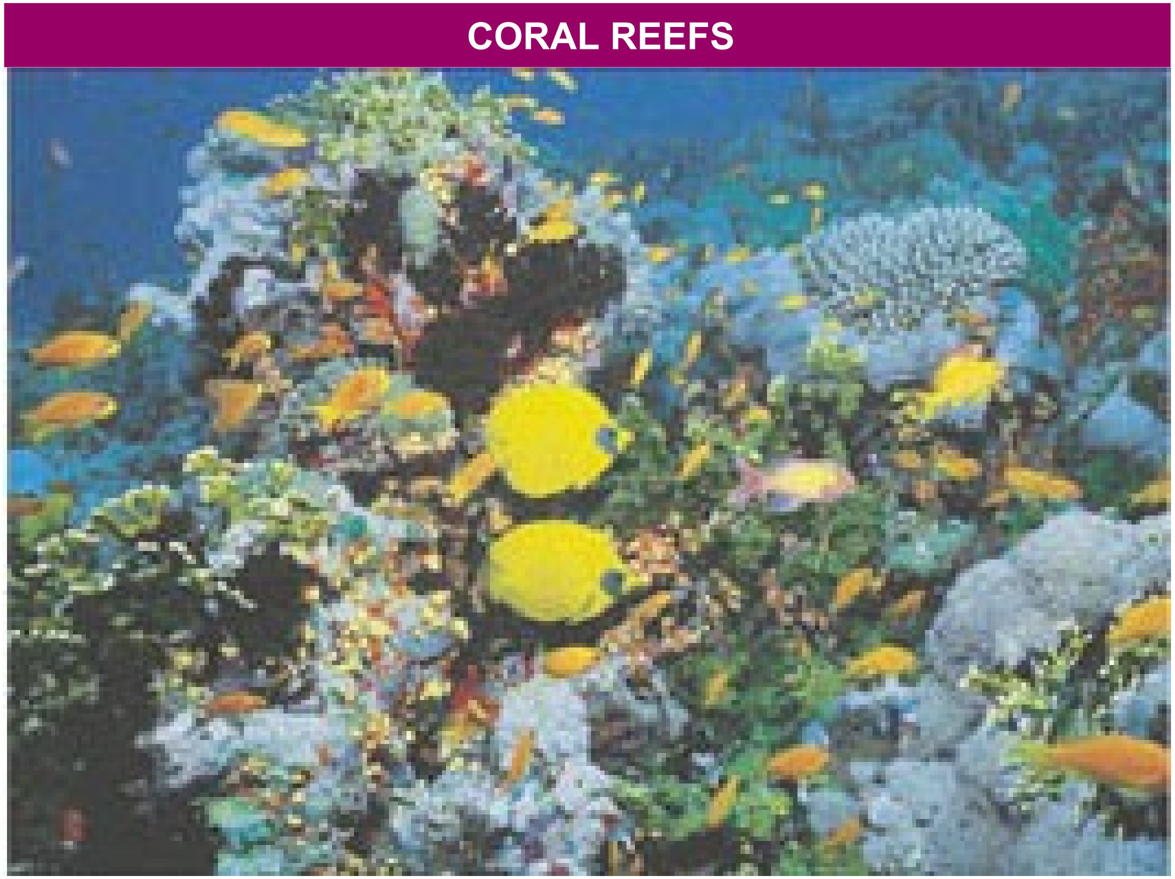


Figure 50.17 A coral reef in the Red Sea

- Marine benthic zone

MARINE BENTHIC ZONE

Figure 50.17 A deep-sea hydrothermal vent community

- Concept 50.4: Climate largely determines the distribution and structure of terrestrial biomes
- Climate
  - Is particularly important in determining why particular terrestrial biomes are found in certain areas

Climate and Terrestrial Biomes

- Climate has a great impact on the distribution of organisms, as seen on a climograph

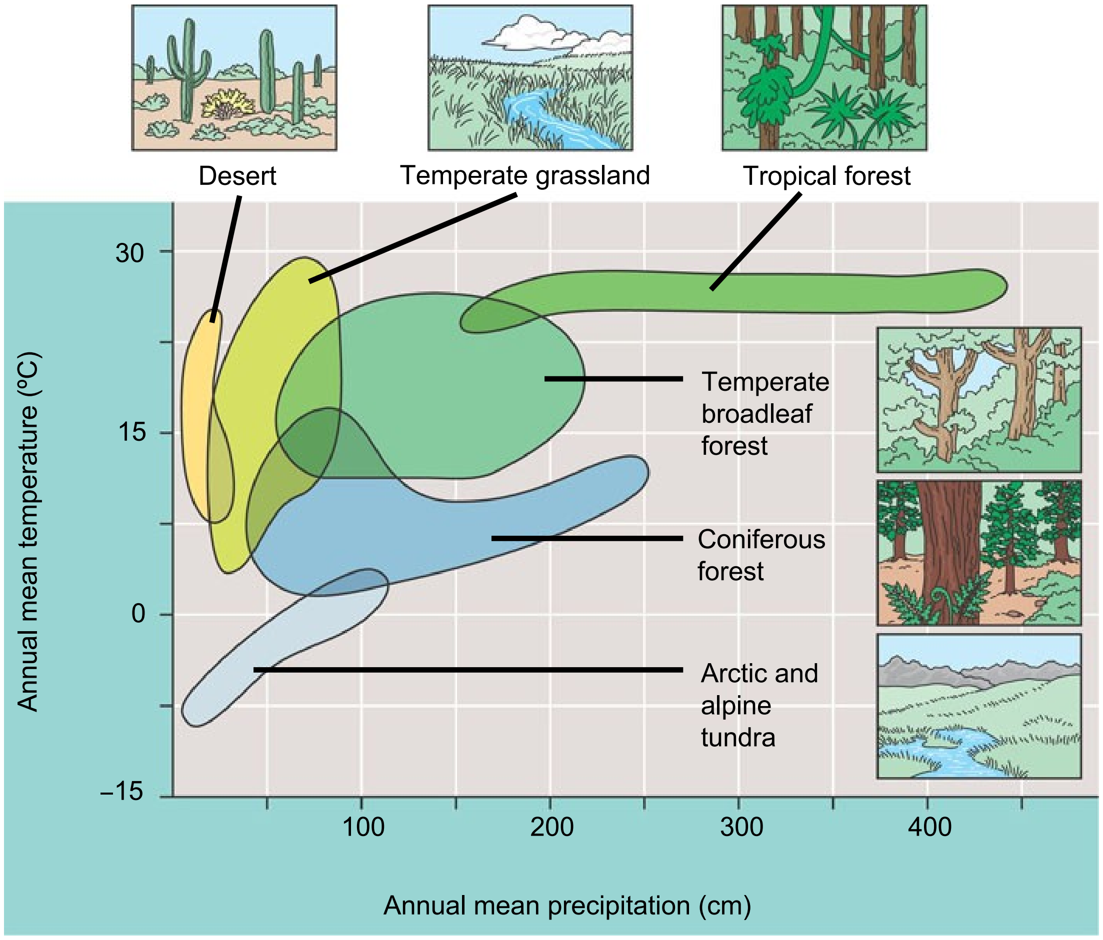


Figure 50.18

- The distribution of major terrestrial biomes

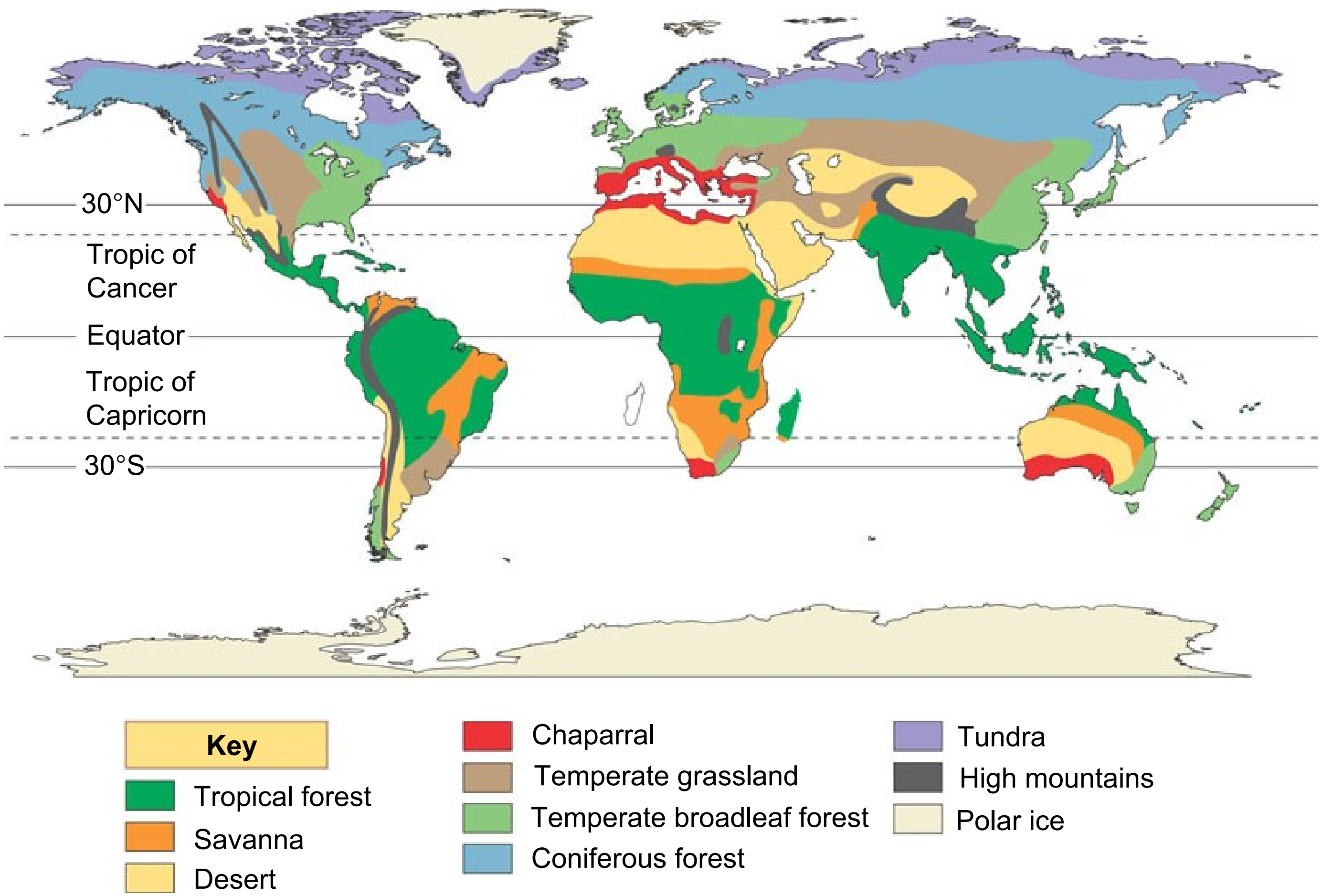


Figure 50.19

General Features of Terrestrial Biomes

- Terrestrial biomes
  - Are often named for major physical or climatic factors and for their predominant vegetation
- Stratification
  - Is an important feature of terrestrial biomes

- Tropical forest

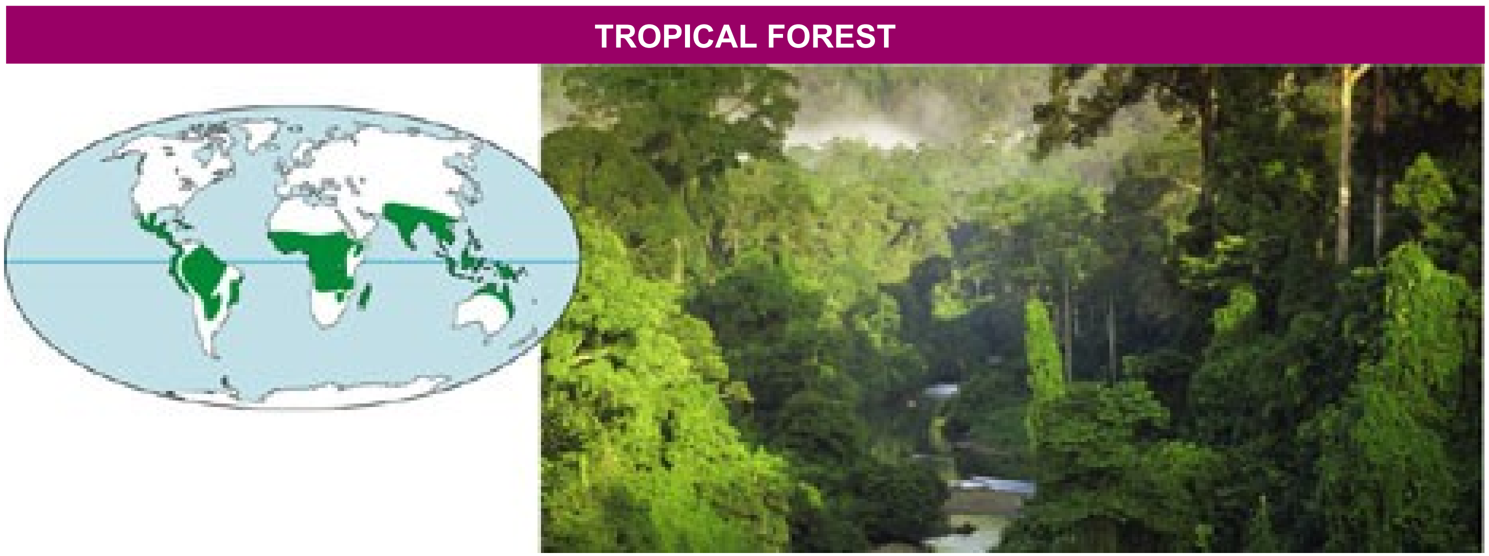


Figure 50.20 A tropical rain forest in Borneo

- Desert

Figure 50.20 The Sonoran Desert in southern Arizona

- Savanna

Figure 50.20

A typical savanna in Kenya

- Chaparral

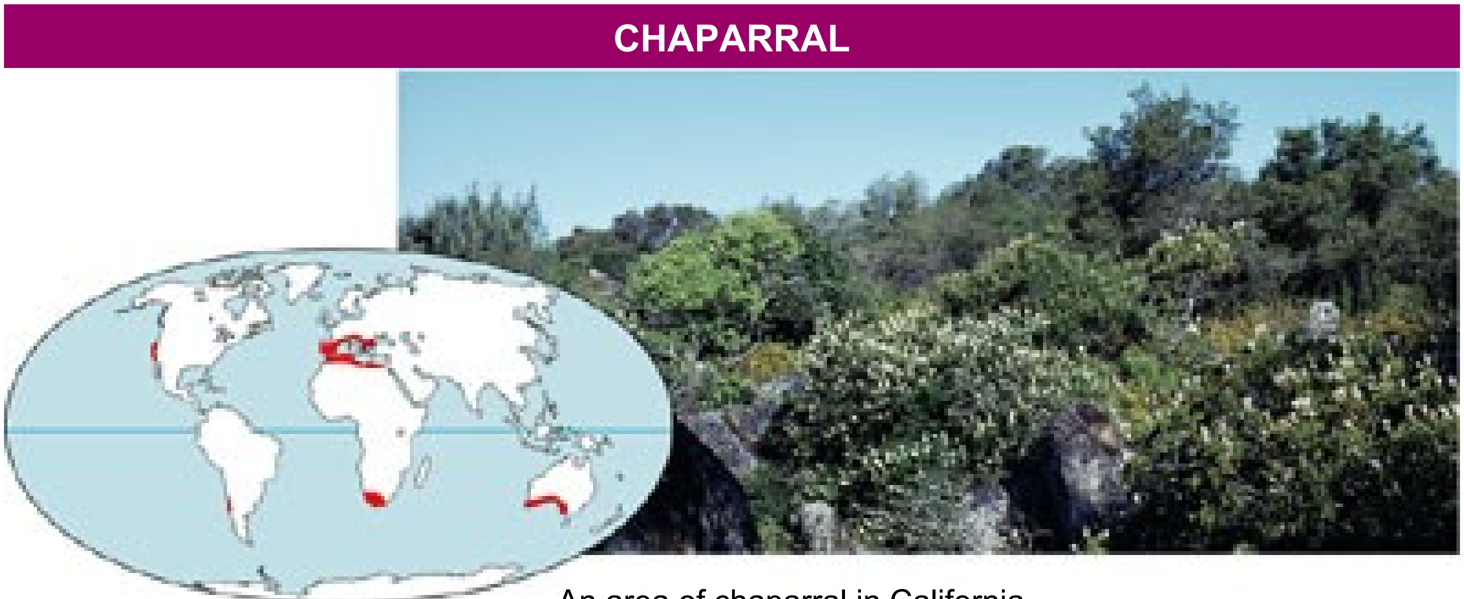


Figure 50.20

An area of chaparral in California

- Temperate grassland

TEMPERATE GRASSLAND

Figure 50.20

Sheyenne National Grassland in North Dakota

- Coniferous forest

Rocky Mountain National Park in Colorado

Figure 50.20

- Temperate broadleaf forest

Figure 50.20

Great Smoky Mountains National Park in North Carolina

- Tundra

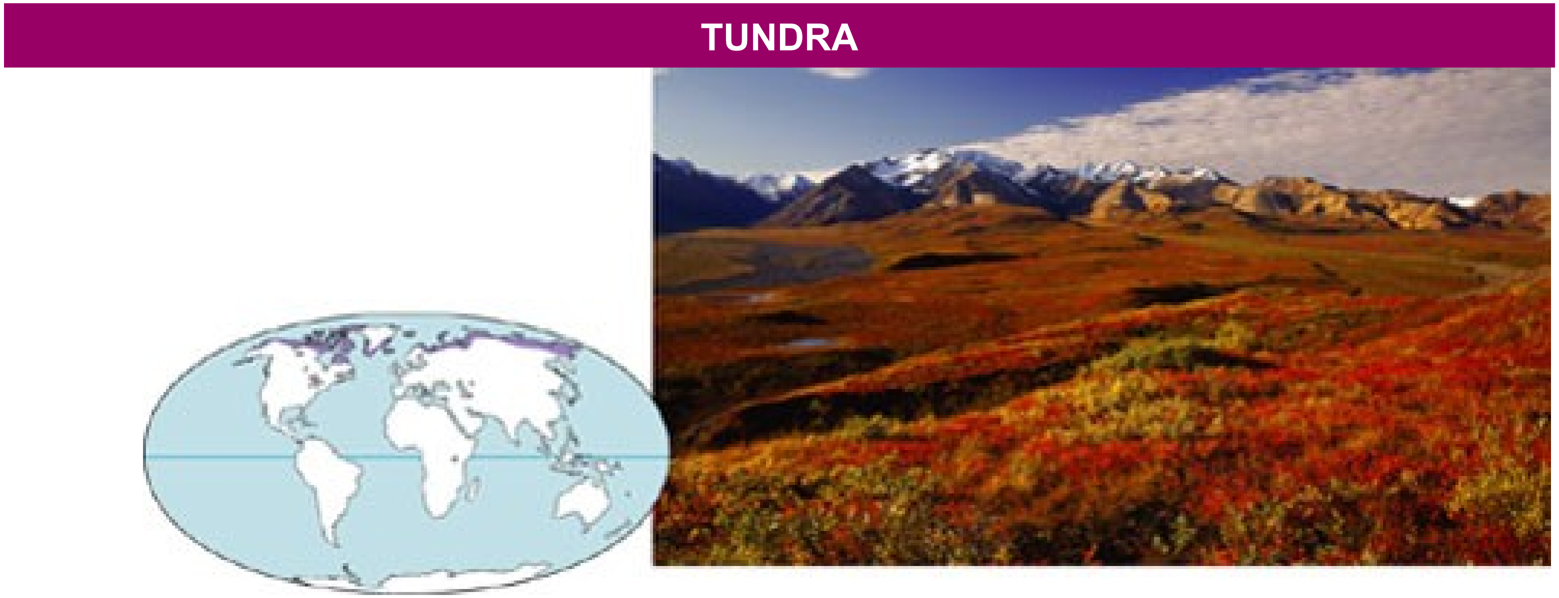


Figure 50.20

Denali National Park, Alaska, in autumn